Project Title: **Project Design Phase-I – Problem/Solution Fit Template**

Full-Stack Stock Practice Trading Simulation Platform

Team ID: STK-2025-01

Define CS, fit into CC

CS

1. CUSTOMER SEGMENT(S)

Who is your customer?

Beginner Investors (18–30)
→ No real trading experience; want to practice with virtual money before using real capital.

Finance Students
→ University or self-study learners testing trading strategies for coursework or growth.

Self-Taught Retail Traders
→ Already read about markets; want a realistic simulation to refine entry/exit strategies.

Finance Educators / Professors
→ Assign virtual trading challenges to students and compare results via leaderboard.

CC

6. CUSTOMER CONSTRAINTS

What prevents customers from taking action or limits their choices?

Financial Risk
→ Fear of losing real money prevents beginners from opening live brokerage accounts.

Knowledge Gap
→ Lack of understanding of order types, fees, and P&L calculations blocks first trades.

Platform Complexity
→ Real trading platforms (Zerodha, Robinhood) are overwhelming for new users.

No Safe Practice Environment
→ Existing simulators use fake/delayed prices and don't replicate real brokerage workflows.

Device & Access Constraints
→ Need web + mobile access; limited broadband for real-time data in some regions.

AS

5. AVAILABLE SOLUTIONS

Existing solutions customers have tried:

Paper Trading (manual)
→ Write down trades on paper. No real-time P&L, error-prone, impossible to scale.

Broker Demo Accounts
→ Zerodha / Robinhood paper mode. Complex UI, no portfolio tracking or history export.

Basic Simulators (Investopedia)
→ Delayed 15-min prices, no watchlist, no notification system, limited mobile support.

Spreadsheets
→ Manual trade logging in Excel. No live quotes, no automated P&L, high maintenance burden.

Our Differentiation
→ Real-time prices + full brokerage workflow + leaderboard + personalised recommendations.

Explore AS, differentiate

Focus on J&P, tap into BE, understand RC

J&P

2. JOBS-TO-BE-DONE / PROBLEMS

Which jobs or problems do you address for your customers?

Practice trading stocks risk-free before going live — build skill and confidence without loss.

Understand portfolio P&L: realised vs unrealised gain, holdings value, and cash balance.

Learn to read price charts and OHLCV data to make informed buy/sell decisions.

Test investment strategies (value, momentum, sector rotation) in a safe environment.

Track full transaction history to review past decisions and identify recurring mistakes.

Compete with peers on a leaderboard to stay motivated and measure skill progress.

RC

9. PROBLEM ROOT CAUSE

What is the real reason this problem exists?

High barrier to entry for real trading:
→ Minimum account balances, KYC verification, and real financial risk deter beginners.

No realistic free simulation exists:
→ Existing simulators use delayed or synthetic prices and lack proper portfolio accounting.

Financial literacy gap:
→ Schools don't teach practical stock trading; most people learn by losing money first.

UX complexity of real platforms:
→ Professional tools prioritise power-users, leaving beginners without a learning path.

Lack of feedback loops:
→ Without P&L tracking and trade history, users cannot learn from past mistakes.

BE

7. BEHAVIOUR

What does your customer do to address the problem and get the job done?

Directly related behaviours:
→ Watch YouTube tutorials on technical analysis
→ Read Investopedia articles on trading basics
→ Open broker demo accounts to see the UI
→ Follow stock subreddits (r/stocks, r/investing)
→ Use Google Finance / Yahoo Finance for quotes

Indirectly related behaviours:
→ Spend hours on financial news (CNBC, Bloomberg)
→ Discuss trades in Discord/Telegram channels
→ Download finance apps (Groww, Moneycontrol)
→ Attend webinars / university finance clubs
→ Track personal spending to build savings for eventual real investment.

Focus on J&P, tap into BE, understand RC

Identify strong TR & EM

TR

3. TRIGGERS

What triggers customers to act?

→ Seeing a friend or peer make profits from stock trading on social media.
→ News about a major IPO or market event (e.g. Tesla rally, crypto crash).
→ University finance course assignment requiring a trading simulation.
→ Reading about compound interest / FIRE movement and wanting to start investing.
→ Job loss or salary cut motivating passive income research via stock markets.

EM

4. EMOTIONS: BEFORE / AFTER

How do customers feel before and after using the platform?

BEFORE: Anxious → Overwhelmed → Confused
→ Scared of losing real money
→ Intimidated by complex brokerage interfaces
→ Unsure where to even start learning

AFTER: Confident → Informed → In Control
→ Comfortable reading charts and placing orders
→ Understands P&L, fees, and portfolio value
→ Ready to open a real brokerage account

SL

10. YOUR SOLUTION

Solution that fits within customer constraints and solves the root cause:

TradeLearning — Full-Stack Stock Simulation Platform

Core capabilities:
→ Virtual account with $10,000 starting balance
→ Real-time stock quotes via Alpha Vantage API (fallback simulated feed if API unavailable)
→ Buy/sell orders with fee calculation & validation
→ Auto-computed portfolio: cash, holdings, P&L
→ Transaction history with date/stock filters
→ Watchlist with optional notes per stock
→ Role-based access: Trader vs Admin
→ Secure JWT auth + HttpOnly refresh cookies
→ Leaderboard ranked by portfolio return %
→ Personalised stock recommendations
→ Achievement badges for learning milestones
→ React web app + REST API (Node.js/Express)
→ MongoDB Atlas — cloud-hosted, replicated

CH

8. CHANNELS OF BEHAVIOUR

8.1 ONLINE

→ Web platform (React SPA on Vercel/Netlify)
→ Mobile-responsive design for phone/tablet
→ Email notifications (trade confirmation, daily P&L summary, performance alerts)
→ Push notifications via mobile browser
→ Social sharing of leaderboard rankings
→ Google OAuth registration / login

8.2 OFFLINE

→ University / classroom assignments using the platform for graded trading challenges
→ Finance club workshops and trading competitions
→ Word-of-mouth referrals between students
→ Printed portfolio reports for course review

Identify strong TR & EM